

Topic 4 - General Vector Spaces

4.1 General vector spaces

Fields

The *scalars* are members of a number system $\mathbb{F}$ called a **field** where addition, multiplication, subtraction and division by a non-zero scalar are defined.

Every field $\mathbb{F}$ contains:

1. A zero scalar s.t. $\alpha + 0 = \alpha$
2. A unit scalar (with $1 \neq 0$) s.t. $1\alpha = \alpha$

Note

Subject scope

MAST10007 mostly looks at $\mathbb{F} = \mathbb{R}$ and $\mathbb{F} = \mathbb{C}$, but also looks at $\mathbb{F} = \mathbb{F}_2 = \{0, 1\}$, the integers modulo 2.

Vector spaces

The aim of vector spaces is to extend the properties of vectors from $\mathbb{R}^n$ to more general sets of objects.

A **vector space** is a set V with two operations defined:

1. vector addition
2. scalar multiplication

We want these two to satisfy the basic properties of vectors in $\mathbb{R}^n$, which leads to a list of 10 *axioms* that give the definition of a vector space.

Definition

Vector space

For a fixed field of scalars $\mathbb{F}$, a **vector space over F** is a non-empty set V with two operations: vector addition and scalar multiplication. The operations are required to satisfy 10 axioms:

Axiom

Additive axioms

Vector addition behaves well:

1. $\mathbf{u} + \mathbf{v} \in V$ (closure in vector addition)
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity)
3. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity)

There must be a zero and an inverse:

1. There exists a vector $\mathbf{0} \in V$ s.t. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$ (existence of zero vector)
2. For all $\mathbf{v} \in V$, there exists a vector $-\mathbf{v}$ s.t. $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ (additive inverse)

Axiom

Multiplicative axioms

Scalar multiplication behaves well:

1. $c\mathbf{v} \in V$ (closure under scalar multiplication)
2. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ (distributivity over vector addition)

- | | |
|-------------------------|--|
| 3. $(c + d)v = cv + dv$ | (distributivity over scalar addition) |
| 4. $(cd)v = c(dv)$ | (compatibility of scalar multiplication) |
| 5. $1v = v$ | (scalar multiplicative identity) |

Examples of vector spaces

Example

Complex numbers $\mathbb{C}$

$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ forms a vector space over $\mathbb{R}$, with addition

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

and scalar multiplication $\lambda(a + bi) = \lambda a + \lambda bi$ for $\lambda \in \mathbb{R}$. The zero vector is $0 + 0i$.

Example

Integers mod 2: $\mathbb{Z}_2^n$

$\mathbb{Z}_2 = \{0, 1\}$ with arithmetic mod 2 is a field, and $\mathbb{Z}_2^n$ forms a vector space over $\mathbb{Z}_2$. Addition is bitwise XOR and scalar multiplication is trivial ($0 \cdot v = \mathbf{0}$, $1 \cdot v = v$). This space is fundamental in coding theory and cryptography.

Example

Polynomials $\mathbb{P}_n$

The set of all polynomials of degree at most n ,

$$\mathbb{P}_n = \{a_0 + a_1x + \dots + a_nx^n \mid a_i \in \mathbb{R}\}$$

forms a vector space over $\mathbb{R}$ under standard polynomial addition and scalar multiplication. The zero vector is the zero polynomial. More generally, $\mathbb{P}$ (polynomials of arbitrary degree) is also a vector space.

Example

Function spaces $\mathcal{F}(X, \mathbb{R})$

The set of all functions $f : X \rightarrow \mathbb{R}$ from a non-empty set X forms a vector space over $\mathbb{R}$, with pointwise operations

$$(f + g)(x) = f(x) + g(x), \quad (cf)(x) = c \cdot f(x)$$

The zero vector is the constant zero function $\mathbf{0}(x) = 0$. Important subspaces include:

- $C[a, b]$: continuous functions on $[a, b]$
- $C^k[a, b]$: k -times continuously differentiable functions
- $C^\infty[a, b]$: smooth (infinitely differentiable) functions

4.2 Subspaces

Definition

Subspace

A **subspace** W of a vector space V is a subset $W \subseteq V$ that is a vector space using the vector addition and scalar multiplication definitions of V .

The subspace theorem can be used to check if a vector space is a subspace (and hence a vector space) without checking all 10 axioms.

Theorem

Subspace theorem

For a vector space V with scalars $\mathbb{F}$, A subset $W \subseteq V$ is a subspace of V iff:

1. W contains the zero vector $\mathbf{0}$ from V
2. W is closed under vector addition ($\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$)
3. W is closed under scalar multiplication ($\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{uv} \in W$)

Example

A plane through the origin in $\mathbb{R}^3$

Let $W = \{(x, y, z) \in \mathbb{R}^3 \mid 2x - y + z = 0\}$. We claim W is a subspace of $\mathbb{R}^3$.

Zero: $2(0) - 0 + 0 = 0$

✓

Closure under addition: Let $\mathbf{u} = (u_1, u_2, u_3), \mathbf{v} = (v_1, v_2, v_3) \in W$, so

$$2u_1 - u_2 + u_3 = 0 \quad \text{and} \quad 2v_1 - v_2 + v_3 = 0.$$

Adding these equations gives $2(u_1 + v_1) - (u_2 + v_2) + (u_3 + v_3) = 0$, so $\mathbf{u} + \mathbf{v} \in W$.

✓

Closure under scalar multiplication: Let $\mathbf{v} = (v_1, v_2, v_3) \in W$ and $c \in \mathbb{R}$. Then

$$2(cv_1) - (cv_2) + (cv_3) = c(2v_1 - v_2 + v_3) = c \cdot 0 = 0$$

so $c\mathbf{v} \in W$.

✓

Hence W is a subspace of $\mathbb{R}^3$.

Example

Symmetric matrices S_n

Let $S_n = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\}$ be the set of $n \times n$ symmetric matrices. We claim S_n is a subspace of $M_{n \times n}(\mathbb{R})$, where the vector space operations are entrywise addition and scalar multiplication.

Zero: The zero matrix $\mathbf{0}$ satisfies $\mathbf{0}^T = \mathbf{0}$, so $\mathbf{0} \in S_n$.

✓

Closure under addition: Let $A, B \in S_n$, so $A^T = A$ and $B^T = B$. Using properties of the transpose,

$$(A + B)^T = A^T + B^T = A + B$$

so $A + B \in S_n$.

✓

Closure under scalar multiplication: Let $A \in S_n$ and $c \in \mathbb{R}$. Then

$$(cA)^T = cA^T = cA$$

so $cA \in S_n$.

✓

Hence S_n is a subspace of $M_{n \times n}(\mathbb{R})$.

Remark

The dimension of S_n is $\binom{n+1}{2} = \frac{n(n+1)}{2}$, since a symmetric matrix is uniquely determined by its entries on and above the main diagonal.

4.3 Linear combinations and spans

Definition

Linear combination

A **linear combination** of vectors $v_1, v_2, \dots, v_k \in V$ is the sum of scalar multiples of the vectors:

$$\sum_{i=1}^k \alpha_i v_i = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k$$

where each $\alpha_i \in \mathbb{F}$.

Definition

Span of a set of vectors

For a vector space V over $\mathbb{F}$, the **span** of vectors $v_1, \dots, v_k \in V$ is the *set of all linear combinations* of the vectors:

$$\text{span}\{v_1, \dots, v_k\} = \{\alpha_1 v_1 + \dots + \alpha_k v_k \mid \alpha_1, \dots, \alpha_k \in \mathbb{F}\}$$

Theorem

Spans and subspaces

For vectors in a vector space, the span of these vectors is a subspace of the vector space.

General method to determine if vectors span $\mathbb{F}^n$

If $A = [v_1 \dots v_k]$ where $v_i \in \mathbb{F}^n$ are column vectors, then:

1. $v_1, \dots, v_k \text{ span } \mathbb{F}^n \Leftrightarrow$ the linear system with augmented matrix $[A \mid w]$ has a solution for all $w \in \mathbb{F}^n$
2. $v_1, \dots, v_k \text{ span } \mathbb{F}^n \Leftrightarrow \text{rank}(A) = n$

Theorem

rank and span

If $k < n$, then vectors $v_1, \dots, v_k \in \mathbb{F}^n$ do not span $\mathbb{F}^n$.

Proof: Since $A = [v_1 \dots v_k]$ has k columns, $\text{rank}(A) \leq k$. Since $k < n$ this implies that $\text{rank}(A) \neq n$. Hence, $v_1, \dots, v_k$ do not span $\mathbb{F}^n$ □

4.4 Linear dependence and linear independence

A vector space can have many sets that span it. To get a small spanning set, we need to remove 'redundant' vectors. This leads to the ideas of **linear dependence and independence**.

Definition

Linear dependence

The vectors $v_1, \dots, v_k \in V$ are **linearly dependent** if there exist scalars $\alpha_1, \dots, \alpha_k \in \mathbb{F}$, with at least one α_i non-zero, s.t.

$$\sum_{i=1}^k \alpha_i v_i = 0$$

.

Definition

Linear independence

The vectors $v_1, \dots, v_k \in V$ are **linearly independent** if there exist scalars $\alpha_1, \dots, \alpha_k \in \mathbb{F}$, s.t.

$$\sum_{i=1}^k \alpha_i v_i = \mathbf{0}$$

has only the solution $\alpha_i = 0$ for all $i < k$.

Theorem

Linear dependence and combinations

The vectors $v_1, \dots, v_k (k \geq 2)$ are *linearly dependent* iff one vector is a linear combination of the others.

Example

Vectors in $\mathbb{R}^3$

Question. Are the vectors

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

linearly independent in $\mathbb{R}^3$?

Solution. Row-reduce the matrix with these as rows:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{matrix} \\ R_3 \leftarrow R_3 - R_1 \\ \end{matrix}$$
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \begin{matrix} \\ \\ R_3 \leftarrow R_3 - R_2 \end{matrix}$$
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{pmatrix}.$$

Three pivots, so v_1, v_2, v_3 are **linearly independent**.

Example

Functions in $C(\mathbb{R})$

Question. Are $f_1(x) = \sin^2 x$, $f_2(x) = \cos^2 x$, $f_3(x) = \cos(2x)$ linearly independent in $C(\mathbb{R})$?

Solution. Suppose $\alpha f_1 + \beta f_2 + \gamma f_3 = 0$ for all x . Evaluating at $x = 0, \frac{\pi}{2}, \frac{\pi}{4}$ gives the system

$$\beta + \gamma = 0, \quad \alpha - \gamma = 0, \quad \alpha + \beta = 0,$$

which has the non-trivial solution $(\alpha, \beta, \gamma) = (1, -1, 1)$. So f_1, f_2, f_3 are **linearly dependent**.

General method for determining if vectors are linearly independent

If $A = [v_1 \dots v_k]$ where $v_i, \dots, v_k \in \mathbb{F}^n$ are column vectors, then:

1. $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{F}^n$ are linearly dependent $\Leftrightarrow$ the linear system with augmented matrix $[A \mid \mathbf{0}]$ has a unique solution.
2. $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{F}^n$ are linearly independent $\Leftrightarrow \text{rank}(A) = k$.
3. If $k = n$ then

$$\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{F}^n \text{ are linearly independent} \Leftrightarrow \det(A) \neq 0$$

Theorem

Vector count

Let $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{F}^n$. If $k > n$, then the vectors are linearly dependent.

Proof: Let $A = [\mathbf{v}_1 \dots \mathbf{v}_k]$ where $\mathbf{v}_i \in \mathbb{F}^n$. Then $\text{rank}(A) \leq n$. Since $k > n$ this implies that $\text{rank}(A) < k$. Hence the vectors are linearly dependent. $\square$

Linear combinations via reduced row-echelon form

Let $A = [\mathbf{v}_1 \dots \mathbf{v}_k]$ and $B = [\mathbf{u}_1 \dots \mathbf{u}_k]$ be matrices with vectors in columns.

If A and B are row-equivalent (ie. $A \sim B$), then the columns of A satisfy the *same linear relations* as the columns of B:

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0} \Leftrightarrow \alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k = \mathbf{0}$$

If $A \sim B$ and B is in RREF, then the relations between the $\mathbf{v}$ vectors can be read off from B.

Example

Row-equivalent matrices share column relations

Let $A = [\mathbf{v}_1 \mid \mathbf{v}_2 \mid \mathbf{v}_3 \mid \mathbf{v}_4]$ where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}.$$

Row-reducing A to RREF gives $A \sim B$:

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 1 & 1 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = B.$$

Columns 1 and 2 are pivot columns; columns 3 and 4 are free. Solving $B\mathbf{x} = \mathbf{0}$ gives $x_1 = x_3 - x_4$ and $x_2 = -x_3 - x_4$, so the null space is spanned by

$$\mathbf{x} = x_3 \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

These two vectors give the **complete** set of linear relations among the columns. By row-equivalence the same relations hold for A:

$$\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}, \quad -\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_4 = \mathbf{0}.$$

We can verify the first directly: $(1, 2, 1) - (2, 3, 1) + (1, 1, 0) = (0, 0, 0)$. $\square$

4.5 Bases and Dimension

Sometimes spanning sets are larger than they need to be, so to get an efficient description of a vector space V , we want to find a set with the *minimum number of vectors* needed to span V .

Definition

Basis

A **basis** for a vector space V is a set of vectors in V which:

1. Spans V , and
2. Is linearly independent.

Note

Unique bases

A vector space V can have many different bases, they are not unique, eg. any two linearly independent vectors in $\mathbb{R}^2$ will span $\mathbb{R}^2$

Theorem

Existence of bases

Let V be a vector space. Then

1. V has a basis.
2. Any set that spans V contains a basis for V
3. Any linearly independent set in V can be extended to a basis of V

Theorem

Length of basis

For a vector space V with basis $\{v_1, \dots, v_k\}$:

1. A subset of V with more than k vectors is linearly dependent
2. A subset of V with fewer than k vectors does not span V
3. Any two bases of V have the same number of elements, k

This allows us to define the dimension of a vector space.

Definition

Dimension

The **dimension** of a vector space V is the number of vectors in any basis for V . This is denoted by $\dim(V)$.

We call V **finite dimensional** if it has a basis with a finite number of elements, and infinite dimensional otherwise.

Note

Empty basis

If $V = \{0\}$, we say that $\dim(V) = 0$ and a basis for V is the empty set.

Standard bases

- The **standard basis for $\mathbb{R}^n$** is

$$\begin{aligned} &\{(1, 0, 0, \dots, 0), \\ &\quad (0, 1, 0, \dots, 0), \\ &\quad \dots, \\ &\quad (0, 0, 0, \dots, 1)\}. \end{aligned}$$

These vectors are typically denoted as $\{e_1, e_2, \dots, e_n\}$. (In $\mathbb{R}^3$ the standard basis is also denoted by $\{i, j, k\}$)

The dimension of $\mathbb{R}^n$ is n .

- The **standard basis for $M_{m,n}$** is

$$\left\{ \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \right\}.$$

The dimension of $M_{m,n}$ is mn .

- The **standard basis for $\mathcal{P}_n$** is

$$\{1, x, x^2, \dots, x^n\}.$$

The dimension of $\mathcal{P}_n$ is $n + 1$.

- Let $\mathcal{P}$ be the set of **all** polynomials of all possible degrees:

$$\begin{aligned} \mathcal{P} = \{ &a_0 + a_1x + \dots + a_nx^n \mid n \in \mathbb{N}, \\ &a_0, a_1, \dots, a_n \in \mathbb{R} \}. \end{aligned}$$

where $\mathbb{N}$ denotes the set of natural numbers. $\mathcal{P}$ is an **infinite dimensional** vector space. The **standard basis for $\mathcal{P}$** is

$$\{1, x, x^2, \dots\}.$$

Theorem

Let V be a vector space, and suppose $\dim(V) = n$ is finite.

1. If a set with exactly n elements spans V , then it is a basis of V
2. If a linearly independent subset of V has exactly n elements, then it is a basis for V

Finding a basis for the span of a set of vectors in $\mathbb{F}^n$

Recall that if $A \sim B$, then the columns of A and B satisfy the same linear relationships.

Method 1: Using columns

To find a basis for $V = \text{span}\{v_1, \dots, v_k\}$ where $v_1 \dots v_k \in \mathbb{F}^n$.

1. Form the matrix A with *columns* $v_1, \dots, v_k$.
2. Reduce A to row-echelon form B .
3. The columns of A corresponding to the leading entries of B give a basis for V .

Note

This method gives a basis that *is a subset* of the original set of vectors.

Method 2: Using rows

To find a basis for $V = \text{span}\{v_1, \dots, v_k\}$ where $v_1 \dots v_k \in \mathbb{F}^n$.

1. Form the matrix A with *rows* $v_1, \dots, v_k$.
2. Reduce A to row-echelon form B .
3. The non-zero rows of B give a basis for V .

Note

This method gives a basis that *usually is not a subset* of the original set of vectors.

This works because elementary row operations do not change the subspace spanned by the rows, and in the row-echelon form the non-zero rows are linearly independent.

4.6 Solution space, column space, row space

Theorem

Solution space

Let $A \in M_{m,n}(\mathbb{F})$ be an $m * n$ matrix with entries in $\mathbb{F}$. Then the solution set

$$S = \{x \in \mathbb{F}^n \mid Ax = \mathbf{0}\}$$

is a subspace of $\mathbb{F}^n$, with $\mathbf{x}$ as a column vector.

This is called the **solution space** or **nullspace** of A , and its dimension is called the *nullity* of A , denoted by $nullity(A)$.

Proof: Follows from the conditions of the subspace theorem. □

Definition

Column space

Let $A \in M_{m,n}(\mathbb{F})$ be an $m * n$ matrix with entries in $\mathbb{F}$. The subspace of $\mathbb{F}^m$ spanned by the columns of A is called the **column space** of A .

Definition

Row space

Let $A \in M_{m,n}(\mathbb{F})$ be an $m * n$ matrix with entries in $\mathbb{F}$. The subspace of $\mathbb{F}^n$ spanned by the rows of A is called the **row space** of A .

Note

- The column space and row space are subspaces, since the span of any set of vectors is a subspace.
- To find a basis for the column space use the column method.
- To find a basis for the row space use the row method.

Effect of row operations

Assume that $A \sim B$ by a sequence of elementary row operations. Then:

- Solution space of A = solution space of B ,
- Row space of A = row space of B ,
- Column space of A is not necessarily the column space of B .

Theorem

For any $m * n$ matrix A :

$$\text{rank}(A) = \dim(\text{row space of } A) = \dim(\text{column space of } A)$$

This is because in both case the number of leading entries in row-echelon form determines the number of basis vectors.

Theorem

Rank-nullity theorem

Suppose A is an $m * n$ matrix. Then:

$$\text{rank}(A) + \text{nullity}(A) = n = \text{number of columns in } A$$

Proof: Suppose A has row-echelon form B . The rank of A = the number of columns in B containing a leading entry. The nullity of A equals the number of columns in B which do not contain a leading entry. Therefore, $\text{rank}(A) + \text{nullity}(A)$ is the number of columns in B , which is the number of columns in A . □

4.7 Coordinates relative to a basis

Theorem

Linear combinations of bases

If $\{v_1, \dots, v_n\}$ is a basis for vector space V , then every vector $v \in V$ can be written *uniquely* in the form

$$v = \alpha_1 v_1 + \dots + \alpha_n v_n$$

Definition

Coordinates

Suppose $B = \{v_1, \dots, v_n\}$ is an *ordered* basis for a vector space V , ie. always arranged in the same order.

$$v = \alpha_1 v_1 + \dots + \alpha_n v_n \text{ for some scalars } \alpha_i \in \mathbb{F}$$

The scalars are uniquely determined by v and are called the **coordinates of v relative to B** .

$$[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$$

is the *coordinate vector of v with respect to B* .

Having a fixed and finite ordered basis B for a vector space V , there is a *one-to-one relationship* between vectors and their coordinate vectors.

Theorem

For a finite dimensional vector space V over a field of scalars $\mathbb{F}$, with B as an ordered basis of V ,

$$[u + v]_B = [u]_B + [v]_B$$

$$[\alpha v]_B = \alpha [v]_B$$

This shows that taking coordinates is a **linear transformation** between V and $\mathbb{F}^n$.